Energy Efficiency of LLM-Generated Code: A Cross-Language and Cross-Platform Empirical Study

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***Abstract*—Evaluations of LLM-generated code focus on correctness rather than the energy the code consumes when executed. We study energy efficiency along three factors: the generating LLM (Claude, GPT, and Gemini via their official coding CLIs), the language (Python, Java, C++), and the hardware platform. 630 programs for 14 tasks are checked for correctness and measured under controlled conditions on an x86 laptop and an Arm Mac. Language dominates: generated Python consumed about 13 times and Java twice the energy of C++. LLMs differ less: in a mixed model, Claude's programs needed 31 % more energy than GPT's, while Gemini's did not differ significantly; the gap stems mainly from algorithmic choices in a few tasks. The LLM ranking did not depend on platform or language. Energy correlated almost perfectly with execution time but hardly with peak memory.**

***Index Terms*—Large language models, code generation, energy efficiency, green software engineering, empirical study**

I. Introduction

Code-generation benchmarks measure correctness, not the energy generated code consumes. Prior studies find that this energy varies between models [1]–[3], but each varies only some factors: several languages without platform interactions [1], [4], or several platforms for Python only [3]. Whether "model A generates more efficient code than model B" holds beyond one language and machine is untested [5].

In a full factorial experiment, three LLMs, accessed through their official coding CLIs, solve 14 tasks in Python, Java, and C++, and every correct program is measured on an x86 laptop and an Arm Mac. We ask which LLM generates the most energy-efficient code (**RQ1**), how languages (**RQ2**) and platforms (**RQ3**) differ, whether the LLM ranking depends on language and platform (**RQ4**), and how energy, time, and memory relate (**RQ5**).

II. Related Work

Most studies measure Python code from several LLMs on one x86 machine [2], [6], [7]. Cursaru et al. [8] use one model on a Raspberry Pi; Apsan et al. [3] use three platforms, but only Python and without platform interactions; Solovyeva et al. [1] cover our three languages on two machines with C++ compiled at `-O0`. Runtime benchmarks such as EffiBench-X [9] omit energy, and for human-written code, language energy rankings largely reflect execution time [10]. No study crosses languages with heterogeneous platforms and models their interactions (Table I).

III. Study Design

a) Design.: LLM, language, and platform are fully crossed; each task–LLM–language cell is sampled five times ($3 \times 3 \times 14 \times 5 = 630$ programs). We measure energy, time, peak memory, and correctness.

b) LLMs.: Flagship models run through their vendors' coding CLIs (Table II), so *LLM* denotes model and tool jointly. Requests run non-interactively in an empty directory with tools disabled or sandboxed and effort *medium*. The same zero-shot prompt (Figure 7) never mentions efficiency; code is used unrepaired.

c) Tasks.: Ten EffiBench-X tasks [9] (one Medium and one Hard per category, drawn randomly) leave the algorithm free; four Computer Language Benchmarks Game (CLBG) tasks prescribe it (Table III). One large input per task is solved R times in-process or sized by N, calibrated so that C++ runs about 1 s on the laptop and no reference exceeds 10 s.

d) Platforms and measurement.: Python 3.14, Java 25, and GCC 14 run natively (Table IV). Energy comes from RAPL [11] (laptop) or Apple's `powermetrics` estimates (Mac); as their scopes differ, platforms are compared only relatively. On the laptop we fix the governor at `performance` and disable turbo; idle power is recorded every 30 min.

e) Procedure and analysis.: On every platform, a program must pass 30 test cases (CLBG: reference output) and match the reference on the large input. Correct programs run three times in random order, including start-up. We fit linear mixed models on log energy with all interactions and random intercepts for task and program, test terms with likelihood-ratio tests, and report Kendall's W and Spearman correlations (**RQ5**).

IV. Results

a) Generation.: All 630 requests returned one code block without tool use; Claude Code answered fastest (Table V).

b) Measurements.: Of 630 programs, 629 were correct on the laptop and 628 on the Mac, where one Java program printed German decimal commas. Repetitions varied by a median 2.6 % (laptop) and 12 % (Mac). Generated Python consumed 11 (Mac: 16) times as much energy as C++, Java 2.5 (2.6) times (geometric means over tasks; Figure 4); human-written EffiBench-X references show a similar Python gap but a larger Java gap (6.5 times; Figure 5). Single algorithmic choices dominate the LLM comparison (Figures 1 and 3): in all

TABLE I
CLOSEST STUDIES (EXT.: EXTERNAL POWER METER).

Study	LLMs	Languages	Platforms	Measurement	Interact.
Cursaru et al. [8]	1	C++, JS, Py	RPi	Ext.	no
Cappendijk et al. [6]	5	Py	1 (x86)	RAPL	no
Islam et al. [2]	20	Py	1 (x86 server)	RAPL	no
Solovyeva et al. [1]	3	Py, Java, C++	Mac (M3), PC (x86)	powermetrics, RAPL	no
Apsan et al. [3]	6	Py	Server, PC, RPi	EnergiBridge, Ext.	partly[a]
Rekhi et al. [4]	10	Py, C++	1 (x86 cloud)	RAPL	no
This study	3[b]	Py, Java, C++	Laptop (x86), Mac (M4)	RAPL, powermetrics	yes

[a] LLM × prompt interaction tested within each platform; no interaction with platform. [b] Accessed via the vendors' official coding CLIs.

TABLE II
LLMS AND THE CLIS USED TO ACCESS THEM.

Vendor	CLI	Model	Identifier
Anthropic	Claude Code	Claude Opus 5.5	`claude-opus-5-5`
OpenAI	Codex CLI	GPT-6.1 Sol	`gpt-6.1-sol`
Google	Antigravity CLI	Gemini 4 Argon	`gemini-4-argon-medium`

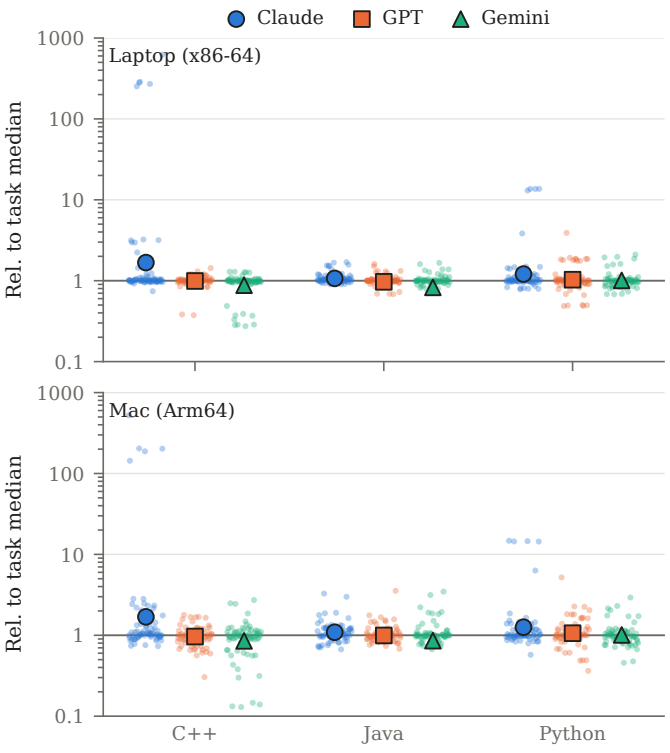


Fig. 1. Program energy relative to its task–language median; large markers: geometric means.

five samples, Claude's C++ solutions for EB02 used 150–300 times the others' energy.

c) Models.: The mixed model (Table VI; per language: Figure 6) confirms effects of the LLM ($p < 0.001$; Claude 1.31 times GPT, Gemini not different) and the language (Java 2.2, Python 12.9 times C++). The LLM ranking depends neither on platform ($p = 0.88$, **RQ4**) nor on language ($p = 0.12$), but

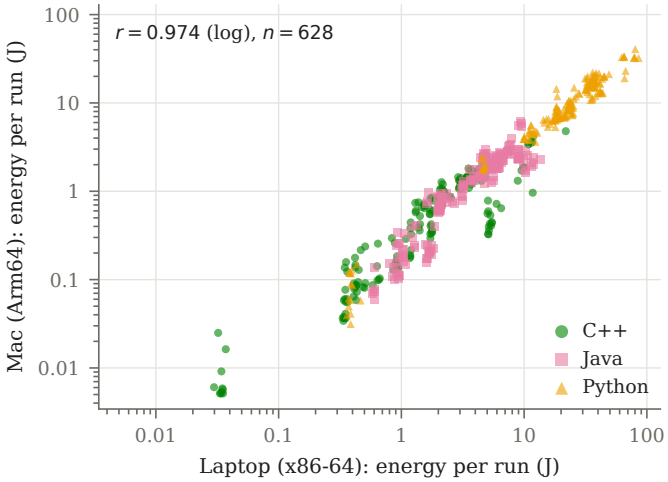


Fig. 2. Energy per program on laptop and Mac (different scopes).

Python costs relatively more on the Mac (language × platform, $p < 0.001$, **RQ3**). Without EB02, Claude's excess shrinks to 1.13 (CI 0.99–1.28). The best LLM per task–language cell agrees across platforms in 31 of 42 cells, yet rankings vary between cells (Kendall's $W = 0.09$ and 0.05). Program energies correlate across platforms ($r = 0.974$, log; Figure 2). Energy follows time ($\rho = 0.99$ on both platforms), as for human-written code [10], but hardly memory ($\rho = 0.24$ and 0.37; **RQ5**).

V. THREATS TO VALIDITY

Energy interfaces differ in scope (`powermetrics` only estimates), so absolute values are not compared across platforms. Hidden system prompts and effort levels make *LLM* a model–tool factor; we record versions and dates. All tasks are public and may be memorised. Some C++ runs last below 0.1 s, where start-up weighs more. With two platforms, results may not generalise to other hardware, models, languages, or real-world code.

VI. CONCLUSION

Language dominates the energy of LLM-generated code: Python needs about 13 times and Java twice the energy of C++. The LLM matters less and mainly through algorithmic

TABLE III
TASKS (EB: EFFIBENCH-X, WITH LEETCODE RELEASE; CLBG: BENCHMARKS GAME).

ID	Task	Stratum / category	Difficulty	Released
EB01	Maximum Total Damage With Spell Casting	Sorting / searching	Medium	2024-06
EB02	Kth Smallest Amount With Single Denomination Combination	Sorting / searching	Hard	2024-04
EB03	Sum of Digit Differences of All Pairs	Mathematics	Medium	2024-05
EB04	Count the Number of Infection Sequences	Mathematics	Hard	2023-12
EB05	Find Subtree Sizes After Changes	String processing	Medium	2024-10
EB06	Palindrome Rearrangement Queries	String processing	Hard	2023-12
EB07	Maximum Score After Applying Operations on a Tree	Dynamic programming / recursion	Medium	2023-11
EB08	Count the Number of Powerful Integers	Dynamic programming / recursion	Hard	2024-01
EB09	Minimum Number of Coins for Fruits	Data structures	Medium	2023-11
EB10	Longest Common Suffix Queries	Data structures	Hard	2024-03
CLBG01	n-body	Floating-point simulation	–	–
CLBG02	spectral-norm	Floating-point linear algebra	–	–
CLBG03	binary-trees	Memory allocation / recursion	–	–
CLBG04	fannkuch-redux	Permutations / integer arithmetic	–	–

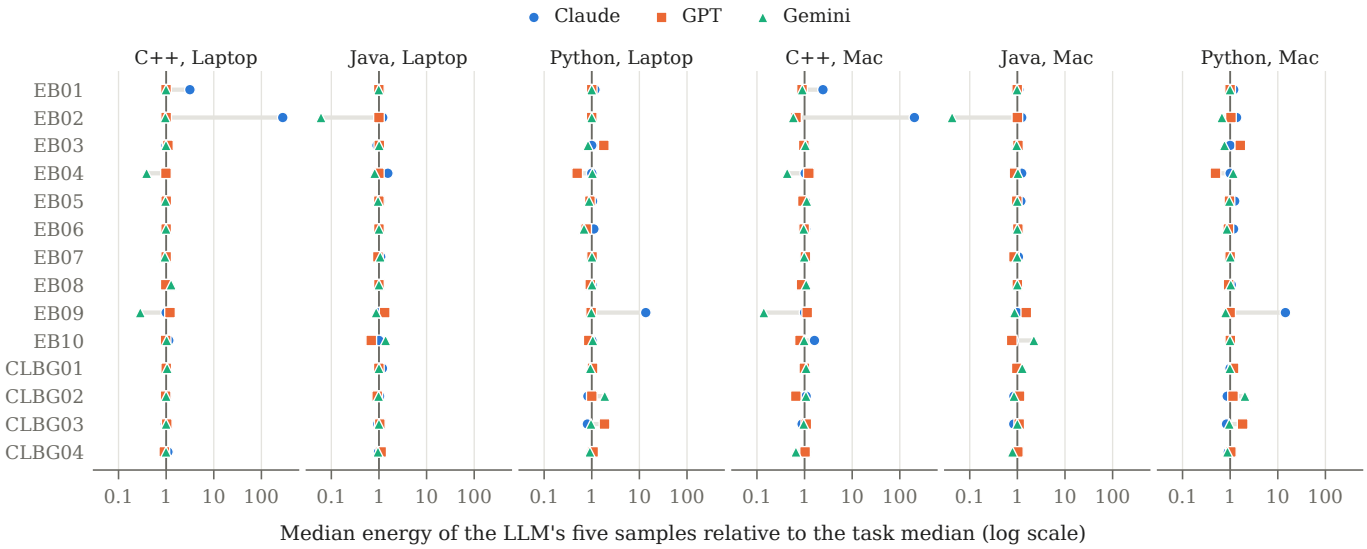


Fig. 3. Median of each LLM's five samples relative to the task median.

TABLE IV
PLATFORMS (C++: GCC WITH `-O2 -std=c++20`).

	Laptop	Mac
CPU / ISA	Intel i7-1185G7 (x86-64)	Apple M4 (Arm64)
RAM	32 GB	16 GB
OS	Ubuntu 24.04, Linux 7.0	macOS 26.3
Python (CPython)	3.14.10	3.14.10
Java (OpenJDK)	25.0.4.1	25.0.5
C++ (GCC)	14.2.0	14.4.0
Energy	RAPL (`powercap`)	`powermetrics` (100 ms)
Scope	CPU package	CPU+GPU+ANE (model)

choices in a few tasks; its ranking holds on x86 and Arm and across languages. As energy follows time, runtime benchmarks remain a useful proxy. More platforms, open-weight models, and energy-aware prompting are future work.

TABLE V
GENERATION TIME PER REQUEST IN SECONDS ($n = 210$ PER LLM).

LLM	Mean	SD	Median	EB mean	CLBG mean
Claude	7.1	3.7	5.9	6.1	9.5
GPT	15.1	5.9	13.8	13.6	18.9
Gemini	17.4	13.8	14.9	14.5	24.6

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TABLE VI
MIXED MODEL ON LOG ENERGY (BOTH PLATFORMS).

	All tasks			Without EB02		
Term	Factor	95% CI	p	Factor	95% CI	p
LLM (baseline GPT)			<0.001			0.028
Claude	1.31	1.10–1.55		1.13	0.99–1.28	
Gemini	0.90	0.76–1.07		0.95	0.84–1.08	
Language (baseline C++)			<0.001			<0.001
Java	2.18	1.84–2.58		1.90	1.67–2.15	
Python	12.93	10.91–15.33		15.20	13.40–17.25	
LLM × language			0.116			0.569
LLM × platform			0.877			0.959
Language × platform			<0.001			<0.001
LLM × language × platform			0.962			0.876

Random intercepts for task and program. Factor: energy relative to the baseline; p: likelihood-ratio test of the term.

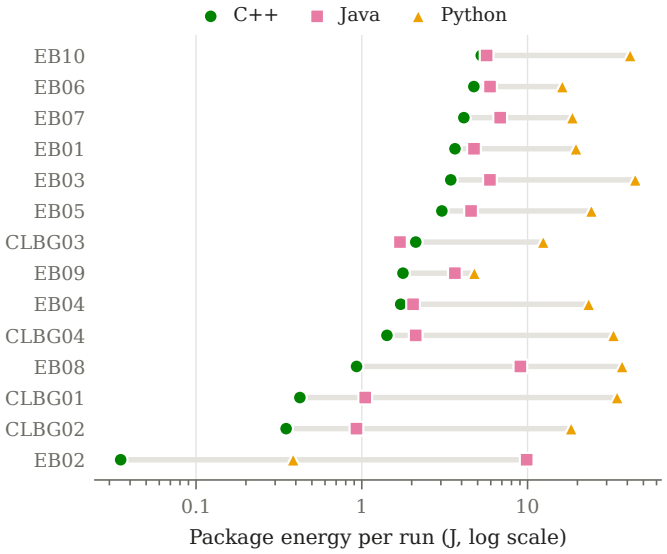


Fig. 4. Median energy per run of generated programs (laptop).

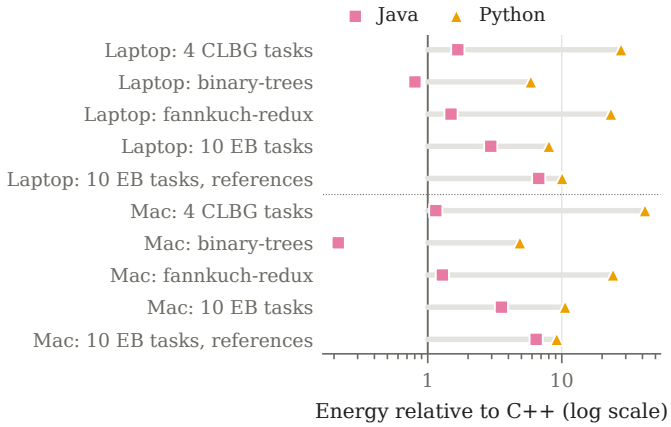


Fig. 5. Java and Python energy relative to C++ (geometric means over tasks).

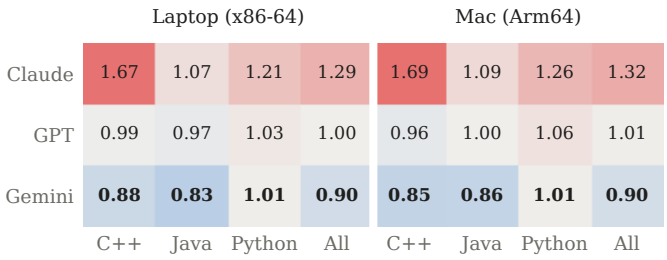


Fig. 6. Geometric mean of program energy relative to the task median; bold: lowest.

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APPENDIX

Figure 7 shows both prompt templates verbatim; they are included directly from the experiment repository (`experiment/prompts/`). Placeholders in braces are filled per task and language: `{language}` is *Python*, *Java*, or *C++*; `{language_note}` is given in Table VII; `{problem}` is the problem statement (EffiBench-X) or the task specification (CLBG); `{starter}` is the EffiBench-X code skeleton for the respective language. Figure 8 shows a fully rendered prompt.

TABLE VII
LANGUAGE-SPECIFIC LINE `{LANGUAGE_NOTE}`.

Language	Text
Python	Target Python 3.14 (CPython).
Java	Target Java 25 (OpenJDK). *Program template only:* The public class must be named `Main`.
C++	Target C++20. The code is compiled with `g++ -O2 -std=c++20`.

(a) Function template (EffiBench-X tasks)

```
Solve the following programming problem in {language}.

Complete the code skeleton given below. Your code will be combined with a test harness
that reads the input, calls your method, and prints the result. Therefore, do not read
from standard input, do not write to standard output, and do not add a main function.

Requirements:
- Return the complete `Solution` class, including any import or include statements it needs.
- Use only the {language} standard library.
- Do not use multiple threads or processes.
- {language_note}
- Respond with exactly one fenced code block containing the code, and nothing else.

Problem:

{problem}

Code skeleton:

```{fence}
{starter}
```
```

(b) Program template (CLBG tasks)

```
Write a complete {language} program that solves the following task.

Requirements:
- The program reads a single integer N from its first command-line argument.
- It writes its result to standard output, exactly in the format specified below.
- Use only the {language} standard library.
- Do not use multiple threads or processes.
- {language_note}
- Respond with exactly one fenced code block containing the code, and nothing else.

Task:

{problem}
```

Fig. 7. Prompt templates used for all LLMs.

````
Solve the following programming problem in C++.

Complete the code skeleton given below. Your code will be combined with a test harness
that reads the input, calls your method, and prints the result. Therefore, do not read
from standard input, do not write to standard output, and do not add a main function.

Requirements:
- Return the complete `Solution` class, including any import or include statements it needs.
- Use only the C++ standard library.
- Do not use multiple threads or processes.
- Target C++20. The code is compiled with `g++ -O2 -std=c++20`.
- Respond with exactly one fenced code block containing the code, and nothing else.

Problem:

You are given an array `nums` consisting of **positive** integers where all integers have the **same** number of digits.

The **digit difference** between two integers is the *count* of different digits that are in the **same** position in
↪  the two integers.

Return the **sum** of the **digit differences** between **all** pairs of integers in `nums`.

**Example 1:**

**Input:** nums = [13,23,12]

**Output:** 4

**Explanation:**
We have the following:
- The digit difference between **1**3 and **2**3 is 1.
- The digit difference between 1**3** and 1**2** is 1.
- The digit difference between **23** and **12** is 2.
So the total sum of digit differences between all pairs of integers is `1 + 1 + 2 = 4`.

**Example 2:**

**Input:** nums = [10,10,10,10]

**Output:** 0

**Explanation:**
All the integers in the array are the same. So the total sum of digit differences between all pairs of integers will be
↪  0.

**Constraints:**

- `2 <= nums.length <= 10^5`
- `1 <= nums[i] < 10^9`
- All integers in `nums` have the same number of digits.

Code skeleton:

```cpp
class Solution {
public:
    long long sumDigitDifferences(vector<int>& nums) {

    }
};
```
````

Fig. 8. Fully rendered prompt for task EB03 (Sum of Digit Differences of All Pairs) in C++.